

الإختصاص والرمز : المعلوماتية – فرع البرمجيات (1010)

المادة : لغة XML

المستندات المسموح بها : لا شيء

المعدل : (10)
المدة : ساعة ونصف

Treat the following exercises

Exercise 1 (5 pts)

It is required to represent an organizer consisting of appointments. For each child element (which is appointment) in the organizer, it is necessary to store the following information:

- “note” consisting of text to explain the appointment
- “date” consisting of year, month, and day. (obligatory)
- “place” consisting of building name and apartment number (the place is optional)
- “time” of the day consisting of hours and minutes (the time is also optional)
- the “time” element contains an obligatory attribute taking one of two values: am or pm.
- the appointment contains an optional attribute representing the name of the person to meet.
 - a) Create a DTD for the organizer.
 - b) Write an XML document representing an organizer for the DTD, containing at least three appointments, and showing all the DTD possibilities (in respect to all optional attributes and elements).

Exercise 2 (5 pts)

Write an equivalent schema for the following fragment of the DTD.

```
<!ELEMENT footballMatch (team, team)>
<!ELEMENT team (clubName, player+, trainer?, previousCup*)>
<!ELEMENT clubName (#PCDATA)>
<!ELEMENT player (name, birthDate, nationality)>
<ATTLIST player shirtNumber (1|2|3) “1”>
<!ELEMENT name (#PCDATA)>
<!ELEMENT birthDate (#PCDATA)>
<!ELEMENT nationality (#PCDATA)>
<!ELEMENT trainer (name, telephone, nationality)>
<!ELEMENT telephone (#PCDATA)>
<!ELEMENT previousCup (#PCDATA)>
<ATTLIST previousCup year NMTOKEN #REQUIRED>
```

N.B: use the types date, integer, string or other types where necessary. The clubName must not exceed 20 characters, and the telephone should be of the format: (99) 9999999

Exercise 3 (4 pts)

Given the following XML file:

```
<letter>
<to>UTBM- Pr. willy</to>
<from>UL-SEK</from>
<subject>indicate the defense date </subject>
</letter>
```

Define a file css to display:

- The element « to » using the following properties:
 - a- display: block
 - b- font-size=12pt
 - c- font-family=Arial
- The element « from » using the following properties:
 - a- display: block
 - b- font in italic
 - c- text-transform: uppercase
 - d- text-decoration: overline

Exercise 4 (6 pts)

Write the XML file as well as the DTD file that correspond to the following schema:

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <!-- definition of simple type elements -->
  <xs:element name="name" type="xs:string"/>
  <xs:element name="name" type="xs:string"/>
  <xs:element name="title" type="xs:string"/>
  <xs:element name="author" type="xs:string"/>
  <!-- definition of attributes -->
  <xs:attribute name="isbn" type="xs:string"/>
  <!-- definition of complex type elements -->
  <xs:element name="topic">
    <xs:complexType>
      <xs:sequence>
        <xs:element ref="name"/>
        <xs:element ref="book"/>
      </xs:sequence>
    </xs:complexType>
  </xs:element>

  <xs:element name="book">
    <xs:complexType>
      <xs:sequence>
        <xs:element ref="title"/>
        <xs:element ref="author"/>
      </xs:sequence>
      <xs:attribute ref="isbn"/>
    </xs:complexType>
  </xs:element>

  <xs:element name="bookstore">
    <xs:complexType>
      <xs:sequence>
        <xs:element ref="name"/>
        <xs:element ref="topic"/>
      </xs:sequence>
    </xs:complexType>
  </xs:element>
</xs:schema>
```